



Bruker **Daltonics**



ultraflex**x**treme

- Redefining MALDI TOF/TOF Performance

think forward

MALDI-TOF/TOF MS

Accelerating Your Applications ...

The new ultrafleXtreme™ leaps the previous limitations of MALDI mass spectrometry technology reaching new levels of data quality. Applications including LC-MALDI Proteomics, high resolution Tissue Imaging based Biomarker Discovery and Top-Down Protein Sequencing are transformed by a multitude of technology innovations and fully integrated proprietary software.

MALDI Imaging

MALDI-Top Down Sequencing

2D-gels

TLC-MALDI

LC-MALDI



PTM analysis

Polymer analysis

... by Superior Technology

True 1,000 Hz MS and MS/MS capability

PAN broadband resolution

smartbeam-II laser

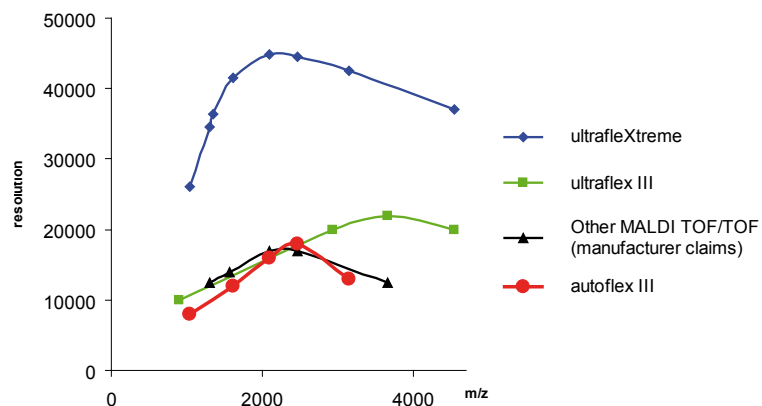
PAC target

FlashDetector™

True kHz systems electronics

MALDI Perpetual ion source

Fully integrated Bruker Software



High mass resolution available

High resolution across a broad mass range provides highest protein identification rates and extensive sequence coverage in bottom-up and top-down analysis.

Zoom into the Proteome

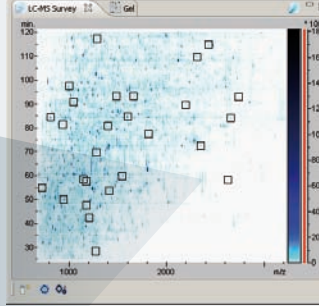
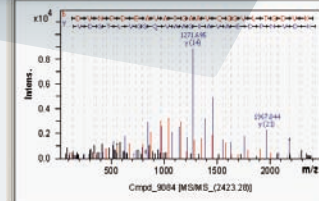
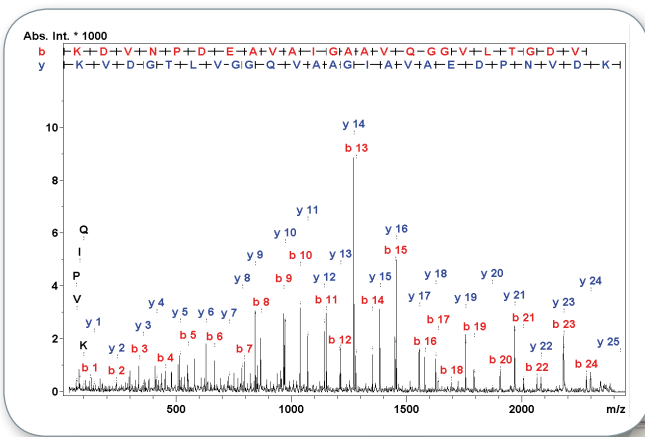
High confidence in protein identification/quantitation

In depth Proteomics analysis is enabled by LC-MALDI workflows on the ultrafleXtreme. Highest confidence of protein identification and reliable quantitation is enabled using any choice of label chemistry (e.g., ICPL-4plex, SILAC or iTRAQ-8plex). ProteinScope 2, the market leading software package allows you to master complex

proteomics data delivering a perfect overview and unsurpassed data archiving capabilities.

Proteomics at your fingertips

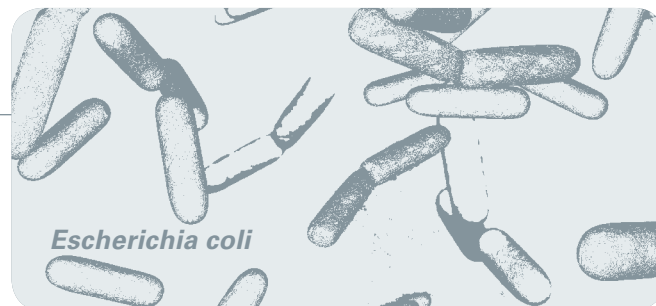
The ultrafleXtreme, in combination with ProteinScope is capable of analyzing 7,000 peptides in MS/MS in less than 8 hours acquisition time. This corresponds to only 4.1 seconds spent on a single MS/MS spectrum. Thus, within nano-LC-MALDI experiments, over 700 proteins were identified (with a FDR at 1 %). In the example shown, ProteinScope organizes all data and provides intuitive and comprehensive data evaluation capabilities.



Row	Protein	MW [kDa]	pI	Peptides	SC	RPMSD [ppm]	Rank
1	Ferredoxin-NADP+ reductase 1 - Escherichia coli (strain K12)	82.1	5.3	29	49.7	5.48	1
2	Formate acetyltransferase 1 - Escherichia coli (strain K12)	25.3	5.6	23	29.1	6.28	2
3	Chaperone protein DnaK - Escherichia coli (strain K12)	69.1	4.7	27	49.4	4.42	3
4	Elongation factor G - Escherichia coli O9:H4 (strain H5)	77.5	5.1	21	34.4	6.44	4
5	Pyruvate dehydrogenase E1 component - Escherichia coli O157:H7	99.6	5.4	28	41.0	6.20	5
6	Chaperone protein DnaJ - Escherichia coli (strain K12)	95.5	5.3	23	34.3	5.23	6
7	ATP synthase subunit beta - Escherichia coli (strain K12)	60.3	4.8	19	57.4	4.88	7
8	60 kDa chaperonin - Escherichia coli O157:H25 (strain E24377A / ETEC)	57.3	4.7	23	45.4	5.68	8
9	2-oxoglutarate dehydrogenase E1 component - Escherichia coli O157:H7	105.0	6.0	20	25.9	6.69	9
10	Tryptophanase - Escherichia coli O157:H7	52.8	5.8	20	46.3	5.48	10
11	DNA-directed RNA polymerase subunit beta' - Escherichia coli O6	195.1	6.7	29	22.9	5.05	11
12	Cysteine synthase A - Escherichia coli (strain K12)	34.5	5.8	16	65.6	4.35	12
13	Aspartate aminotransferase - Escherichia coli (strain K12)	52.3	5.1	18	36.0	6.10	13
14	Outer membrane protein A precursor - Escherichia coli (strain K12)	37.2	6.0	15	95.8	5.26	14
15	Succinyl-CoA synthetase beta chain - Escherichia coli (strain K12)	41.4	5.2	18	53.9	4.90	15
16	Peptide: dipeptide-binding protein precursor - Escherichia coli (strain K12)	60.0	6.0	20	50.3	6.52	16
17	Elongation factor Tu 1 OS-Escherichia coli O157:H25 (strain E24377A / ...)	43.3	5.2	17	38.8	5.27	17

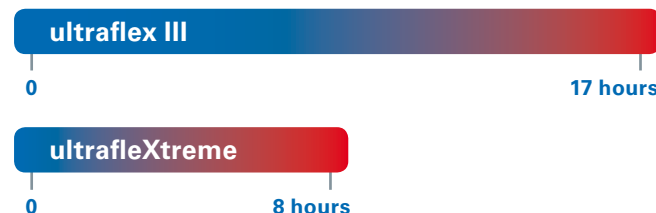
ProteinScope 2 is ideal for display and validation of LC-MALDI experiments: single run nano-LC-MALDI analysis of 500 ng E.coli lysate with 707 identified proteins.

- **Speed & Sensitivity for Demanding Biomarker Applications**



LC-MALDI at full speed, always

Outstanding performance at even shorter acquisition times is achieved with the ultrafleXtreme. The patented MALDI Perpetual™ ion source self-cleans with an IR-laser beam in just a few minutes to ensure maximum sensitivity from each LC-MALDI run.

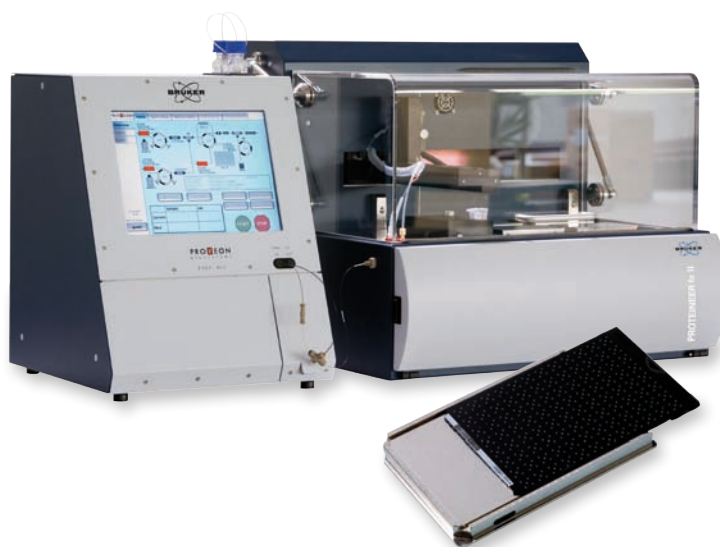


Full Bruker system integration

The splitless EASY-nLC™ system with minimal solvent consumption and the PROTEINEER fc II™ fraction collector are the perfect LC-MALDI companions. Fully integrated in the Compass software, they provide robust and reliable LC-MALDI sample preparation.

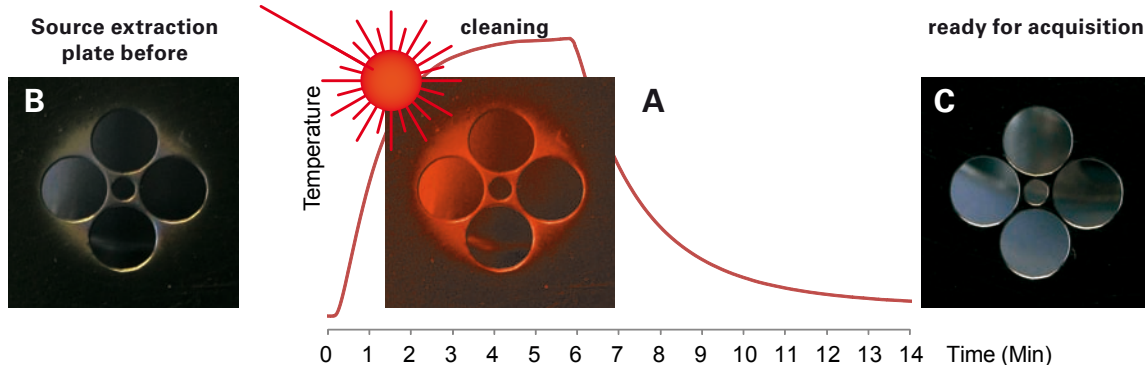
Proprietary Prespotted AnchorChip™ (PAC) targets increase the achievable sensitivity by another factor of 10 and eliminate the need to add matrix or calibrants. Single-use PAC targets are memory-effect free for maximum reliability of protein identification and quantification.

500 ng E.coli lysate: >700 Proteins are identified (FDR 1%) on ultrafleXtreme, at further reduced acquisition time for LC-MALDI analysis.



Read more: Tuerk et al. 2009 Tracking and quantification of (32)P-labeled phosphopeptides in LC-MALDI MS. Anal. Biochem. PMID: 19376078

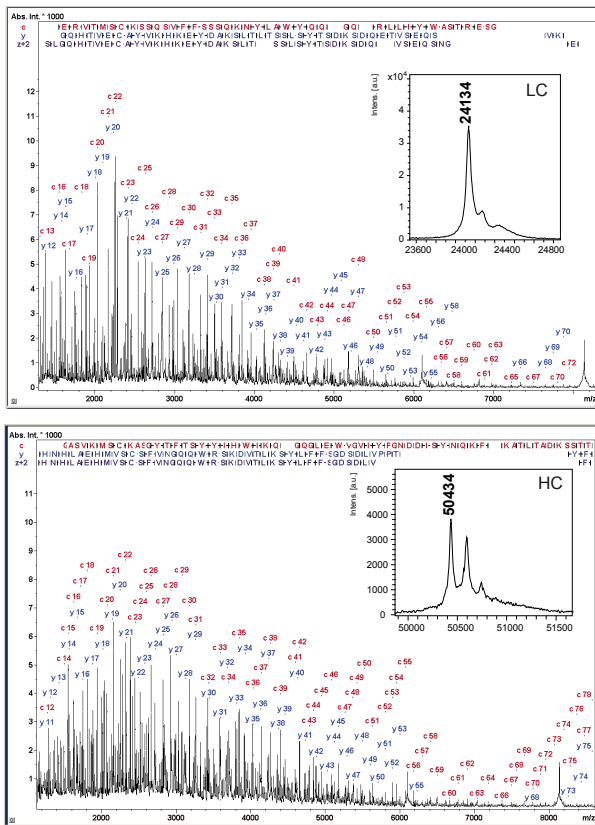
Self-cleaning ion source based on laser-irradiation



MALDI Perpetual Ion Source: The proprietary MALDI Perpetual ion source is automatically laser-cleaned in minutes. The unique, laser-based cleaning process removes all MALDI matrices without residues. Extraction plate temperature profile during the cleaning process (A). B. Extraction plate before and (C) after the laser cleaning process.

Highest Resolution for Unsurpassed Data Quality

Sequencing of immunoglobulins



MALDI Top-Down Sequencing analysis of reduced IgG light (LC) and heavy chain (HC). N-terminal HC pyroglutamylation is revealed as well as the glycosylation heterogeneity. More than 130 residues can be read out from the terminal sequences with isotopic resolution maintained.

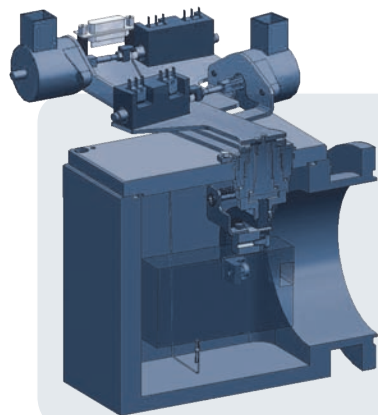
Top-down sequencing with highest sequence coverage

The ultrafleXtreme provides unmatched Top-Down Sequencing (TDS) capabilities: direct sequencing and identification of intact, undigested proteins. Assignment of N- and C-terminal sequences and modifications such as Glycosylation and PEGylation, can also be analyzed on the ultrafleXtreme.

MALDI-TDS is a Next Generation Protein Sequencing technology for Biotherapeutics, Biologics and Biosimilars. Further application areas are protein isoform differentiation, domain mapping and the monitoring of *in vivo* and *in vitro* proteolytic processing events.

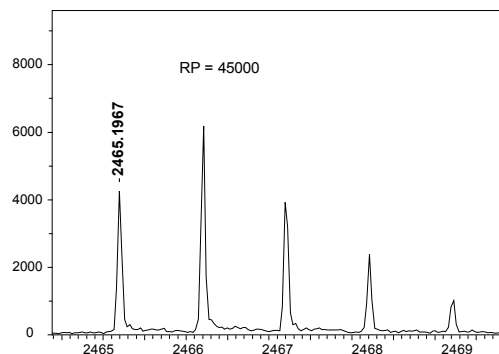
Dedicated Top-Down Sequencing (TDS) expert tools implemented in the BioTools 3.2 software enable the facile readout of TDS information from automated spectra acquisitions.

Read more: Application Note MT-96: Calling N- and C-terminal Protein Sequences with High Confidence and Speed: MALDI-TDS applied to the ABRF-ESRG 2009 Research Study.



FlashDetector

ultrafleXtreme delivers mass accuracy < 1 ppm and superb resolving power >40,000 simultaneously across the Proteomics mass range due to its patented PAN™ technology and unique FlashDetector.



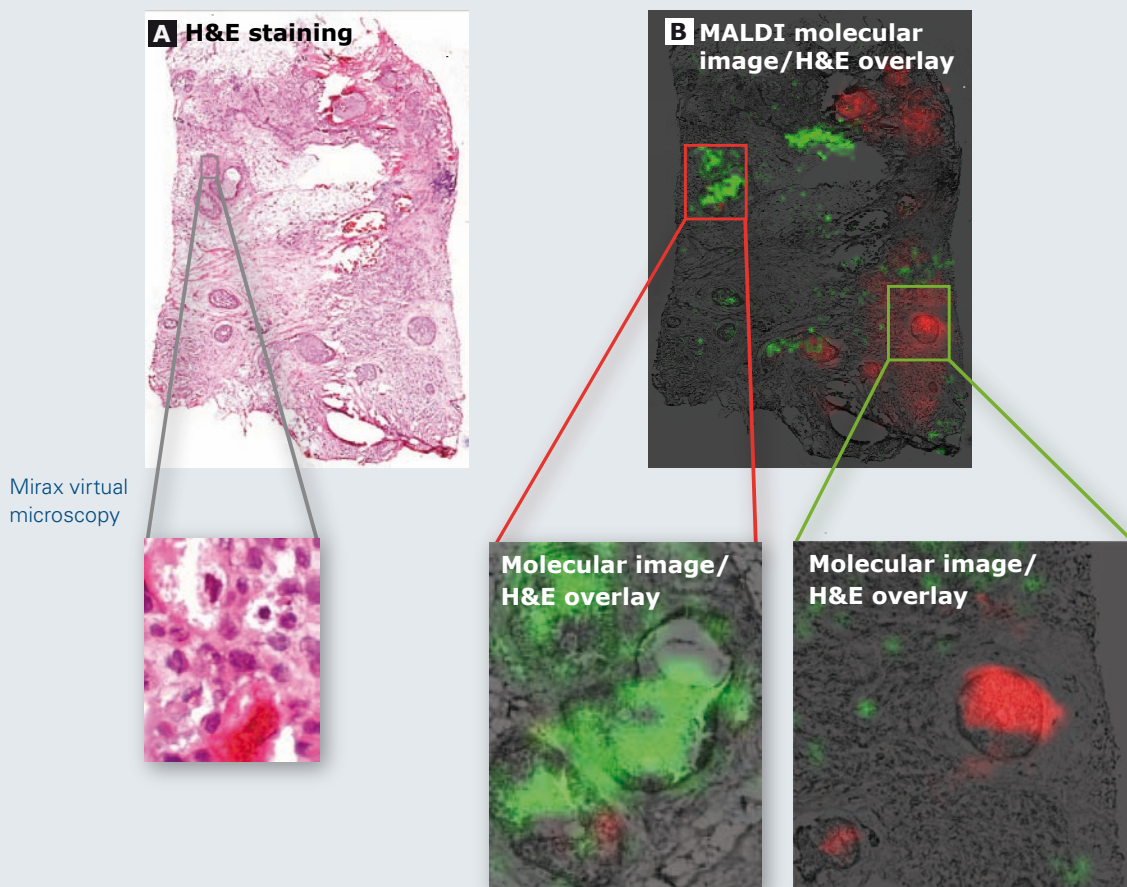
● Molecular Histology

Hyphenating molecular histology and virtual microscopy

A new integrated solution provides full-bandwidth molecular information to histology. The system allows direct overlay of the full microscopic image with the full molecular and spectral information in one convenient visualization software tool, enabling true molecular histology. Proprietary technologies such as the smartbeam-II™ laser for best spectra quality at highest speed and best spatial

resolution, as well as the Bruker ImagePrep™ for easy and reproducible sample preparation and fully integrated flexImaging software, have turned MALDI imaging into a robust, reproducible and easy to use research application.

Now, the full integration of Mirax Scan virtual slides (Zeiss) allows pathologists to evaluate their results directly in the histological context with the full spatial resolution of optical microscopy and the molecular information from MALDI imaging.



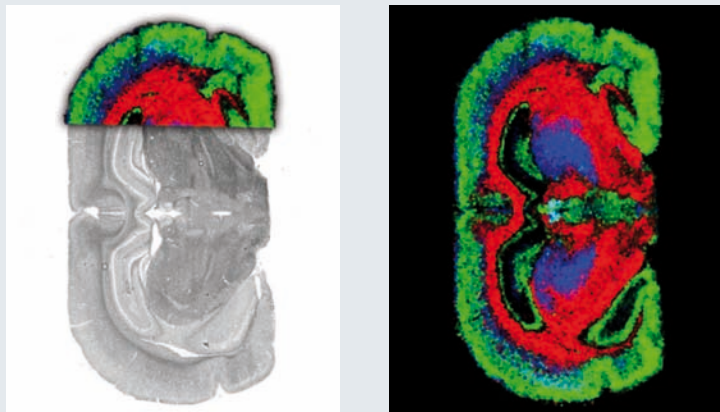
Dual mode optical (A) and MALDI tissue imaging (B) of a human breast cancer biopsy. Several ductal carcinoma in situ (CIS) are present in the tissue. Two molecular signals (shown in red and in green) allow the detections of CIS sub-groups and precise co localization of the CIS in both image representations (bottom: center and right). The integrated Mirax (Zeiss) virtual microscopy permits direct evaluation of these findings on the morphological level. Sample courtesy of Dr. Axel Walch, Helmholtz-Centre Munich, Germany

● MALDI Imaging at High Resolution and Speed

Prerequisite for high resolution imaging is a stable, robust laser operation with long lifetime, a requirement mastered by the Bruker smartbeam-II laser. In addition, a true 1 kHz system performance all the way through the data handling pipeline is required to make use of the kHz laser in a true molecular imager with unmatched performance.

The unique smartbeam-II laser with 10 μm laser focus size provides molecular tissue images near single cell resolution.

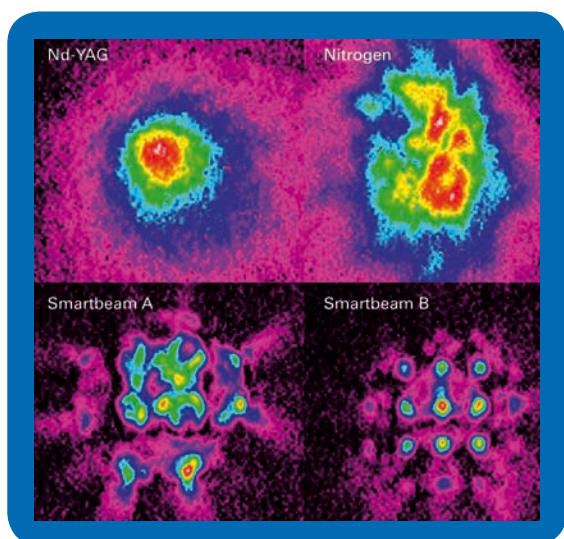
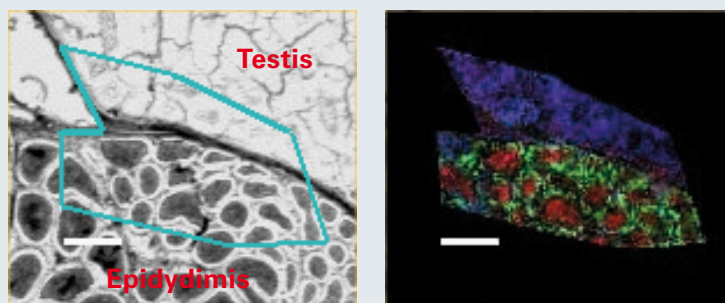
The speed advantage of 1 kHz imaging



One kHz acquisition on ultrafleXtreme (right) with 2 pixel/sec vs. classical 200 Hz acquisition (left) result. Sample: 10 μm mouse brain section. MALDI image overlaid on scanned tissue image after identical acquisition time.

Lipid imaging of rat testis/epididymis. Lateral resolution 20 μm , scale bar: 500 μm . The selected lipids show a specific distribution in testis (blue: m/z 782.5), in the walls of the tubuli (green) and the spermatozoa (red) in the epididymis. Left: optical image with marked acquisition region.

Lipid imaging

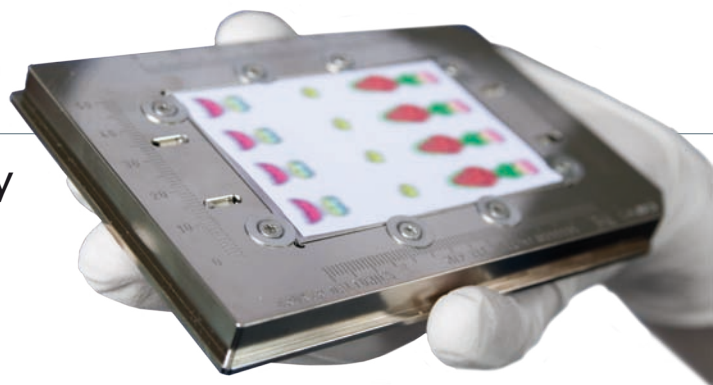


smartbeam-II

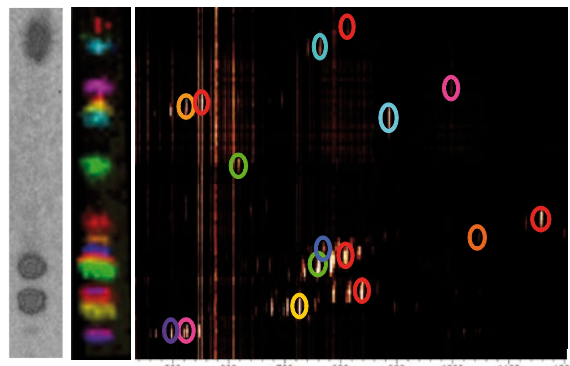
The further development of smartbeam leads to a dramatically increased performance. Repetition rates of up to 1,000 Hz can now be achieved with modulated beam profile (A, B). The smartbeam enables high numbers of shots on identical spot positions of 30 MS/MS spectra and more.

Read more: Holle et al. (2006).
Optimizing UV laser focus profiles for improved MALDI performance
J. Mass Spectrom. 2006; 41: 705–716

● Beyond Proteomics: Lipid Analysis Made Fast & Easy



Direct analysis from TLC-plates

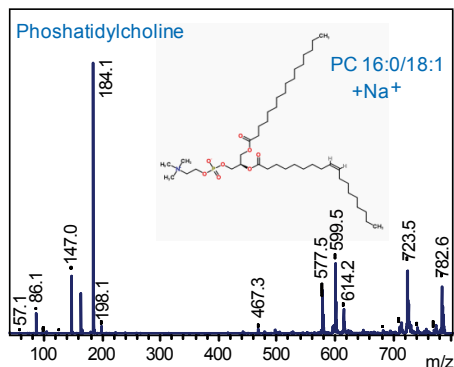


Direct analysis from TLC-plates!

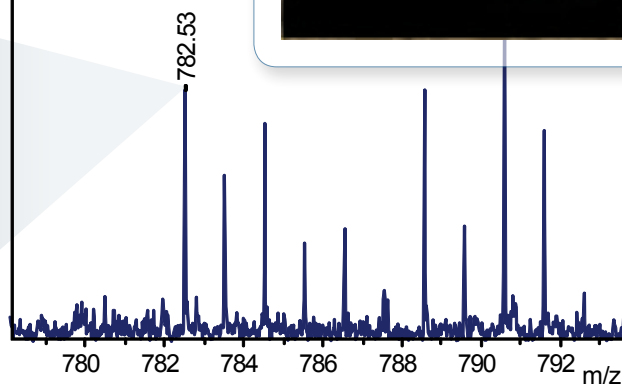
TLC-MALDI is Bruker's proprietary solution for the direct analysis of samples from thin layer chromatography (TLC). TLC-MALDI detection enables high sensitivity read-out of separation traces directly from the TLC plate.

Lipid ID directly from MALDI image

The lipid at m/z 782.5 (blue) was selected for MS/MS and phosphatidylcholine 16:0/18:1 was identified right from the image.



PC 16:0/18:1
+Na⁺



Reliable sample prep

The ImagePrep coats tissue surfaces and TLC plates with a fine mist of 20 μ m matrix droplets with optical active feedback control. Coating is applied in several cycles for maximum signal and preserves the analyte localization for outstanding image resolution in a fully automated process.

Technical Specifications

Redefining MALDI TOF/TOF Performance

- 1 kHz smartbeam-II™ laser technology enables ultra-high data acquisition speed in both MS and MS/MS-mode.
- Laser focus diameters down to 10 µm for high spatial resolution imaging without pixel overlap.
- Wide mass range resolving power up to 40,000 due to proprietary PANTM technology.
- FlashDetector™ combined with new 4 GHz digitizer provides 1 ppm mass accuracy for highest confidence.
- MALDI Perpetual™ Ion Source with laser-based self-cleaning in minutes. It's push-button concept for removing sample and matrix debris is indispensable for kHz operation.

Compass software suite

- flexControl for quick and easy instrument control.
- flexAnalysis for automatic and interactive data analysis.

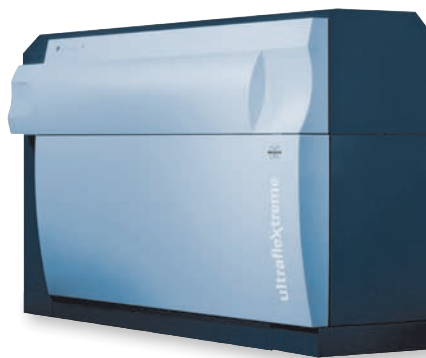
Optional bioinformatics packages

- BioTools™, the cutting edge software for protein analysis and Top-Down sequencing
- WARP-LC™, the LC-MALDI and protein quantitation software.
- flexImaging™, the leading platform for MALDI Imaging.
- PolyTools™, interactive interpretation of MALDI polymer spectra.
- ProteinScape™ for comprehensive handling of proteomics projects.
- Compass Security Pack™: assisting in 21CFR part 11 compliance.

Support features

- Extended self diagnostics
- Remote on-line service and support capability.
- IQ/OQ/PV procedures for regulated environments.
- Maintenance contracts in various levels available.

For research use only.
Not for use in diagnostic procedures.



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